SOP Number and version: SOP-NAT-CULT.v01

SOP Title: **Bacteria Cultivation**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
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Approval	Frank J. Hernandez	Principal investigator		09/01/2025

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Approved by PI (Frank J. Hernandez) - 2026-02-17T12:32:00.119Z

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1. Overview and background

Cultivating bacteria is crucial for further testing such as identification, enumeration and performing susceptibility testing. Sample types, inoculation methods and media can vary based on the purpose and/or the type of bacteria being used/expected to grow.

2. Scope

In this SOP, bacteria cultivation methods used in Nucleic Acids Technology laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) will be explained.

This SOP covers the sections below.

- Media preparation for cultivating bacteria
- Inoculation techniques
- Storage of bacteria

3. Procedure

3.1. Media preparation

3.1.1. Liquid media preparation

- To understand the right proportions of the media powder and recommended solvent, carefully read the instructions of the product provided by the manufacturer.

- Use a high precision analytical lab scale **(Figure 1)** and plastic weighing boats **(Figure 2)** for weighing the commercially obtained powder formulation of desired media (e.g., Tryptone Soya Broth).
- Mix the powder with the solvent (e.g., deionized water) recommended by the manufacturer in 500-1000 mL glass bottles.
- Shake the bottle well until the powder media dissolves.
- Autoclave the media as described in **SOP-NAT-STER**.
- After the sterilization, wait until the media reaches to room temperature
- Keep the media at 4ºC until it ends (discard the media when %80 is consumed) or expires (manufacturer`s recommendation).

3.1.2. Agar media preparation

- To understand the right proportions of the media powder and recommended solvent, carefully read the instructions of the product provided by the manufacturer.
- Use a high precision analytical lab scale **(Figure 1)** and plastic weighing boats **(Figure 2)** for weighing the commercially obtained powder formulation of desired media (e.g., Tryptone Soya Broth).
- Mix the powder with the solvent (e.g., deionized water) recommended by the manufacturer in 500-1000 mL glass bottles **(Figure 3).**
- Shake the bottle well until the powder media dissolves.
- Autoclave the media as described in **SOP-NAT-STER**.
- When autoclaving process is ongoing, place sterile 90 mm petri dishes **(Figure 4)** on the media preparation bench that is previously disinfected with alcohol as described in **SOP-NAT-STER**.
- After autoclaving, pour the recommended volume of media into the petri dishes when it is still warm using an autoclave glove to prevent skin damage **(Figure 5).**
- Wait until the media solidifies at room temperature.
- Place the plates into refrigerator and keep at 4ºC until use.

Figure 1. Lab scale

Figure 2. Weighing boats

Figure 3. Glass media bottles

Figure 4. Petri dishes

Figure 5. Autoclave glove

3.2. Inoculation techniques

3.2.1. Agar culture

3.2.1.1. Quadrant streak method

This inoculation technique is the primary method utilized at Nucleic Acids Technology laboratories, except when bacterial enumeration in a sample is required. It is applicable for inoculating various sample types, including colonies from agar media, storage beads, clinical specimens, and liquid cultures. The primary advantage of this technique is its ability to isolate single colonies, even from samples containing multiple bacterial types, facilitating further analysis and characterization **(Figure7)**.

- Place the sample (e.g., a storage bead or 10 µL of a liquid sample) close to the edge of the plate in sterile conditions.
- Using a sterile inoculation loop, spread the sample in a small area (≈1cm^2) on the edge.
- Streak the sample from that edge to another crossing entire plate.
- Go back to the starting point where the sample has been initially placed and start making zig-zag inoculation streaks from edge to edge till a quarter of the plate (**1st inoculation area**)
- Sterilize the inoculation loop.
- Starting from the right corner of the 1st inoculation area (treat inoculations streaks in this area as sample), make zig-zag inoculation streaks touching the “sample” three times and continue until another quarter of the plate is covered (**2nd inoculation area**).
- Sterilize the inoculation loop.
- Starting from the right corner of the 2nd inoculation area (treat inoculations streaks in this area as sample), make zig-zag inoculation streaks touching the “sample” three times and continue until another quarter of the plate is covered (**3rd inoculation area**).
- Sterilize the inoculation loop.
- Starting from the right corner of the 3rd inoculation area (treat inoculations streaks in this area as sample), make zig-zag inoculation streaks touching the “sample” three times and continue until another quarter of the plate is covered (**4th inoculation area**).

- Incubate the agar plate at 36±1°C for 18-24 hours.

Always consider specific atmospheric conditions that the bacteria need to grow. Use an anaerobic jar system and pouches for the desired atmospheric conditions, if needed, according to the manufacturer`s recommendations.

Figure 6. Anaerobic jar system and gas pouches for specific atmospheric conditions.

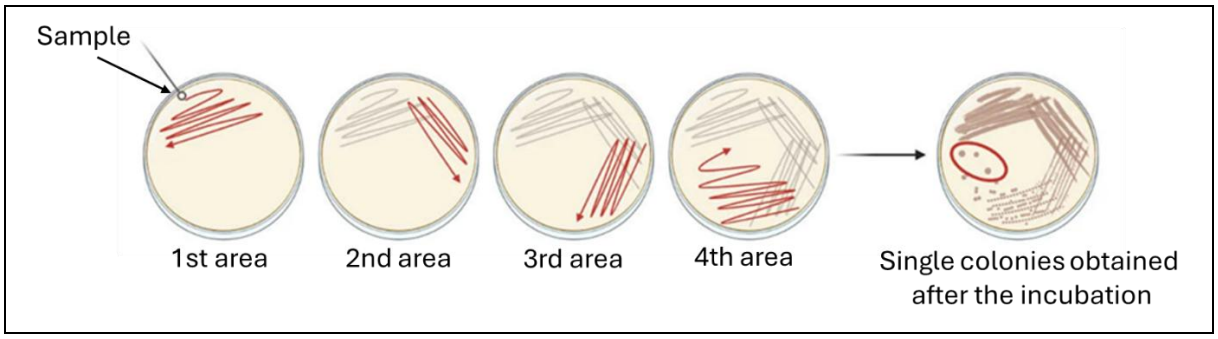

Figure 7. Steps of the quadrant streak method. Inoculation loop must be sterilized between each step.

3.2.1.2. Semi-quantitative continuous streak method

This method is used for enumeration of bacteria in a sample (colony counting) as described in **SOP-NAT-CFU.** To obtain accurate results representing the whole sample, make sure that the sample is homogenous before the inoculation, so that any part of it that can be used for inoculation has the same number of bacteria **(Figure 8).**

- Place 10 µL sample close to the edge of the plate in sterile conditions.
- Using a sterile inoculation loop, spread the sample in a small area (≈1cm^2) on the edge.
- Streak the sample from that edge to another crossing entire plate.
- Go back to the starting point where the sample has been initially placed and start making zig-zag inoculation streaks from edge to edge till the end of the plate.
- Incubate the agar plate at 36±1°C for 18-24 hours.

- Always consider specific atmospheric conditions that the bacteria need to grow. Use an anaerobic jar system and pouches for the desired atmospheric conditions, if needed, according to the manufacturer`s recommendations.

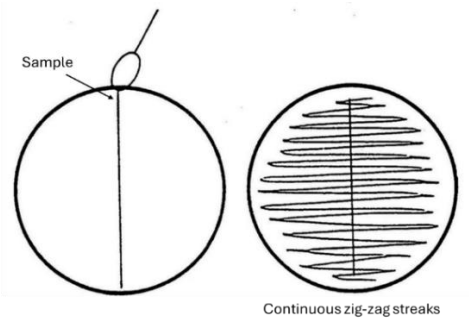

Figure 8. Steps of the semi-quntitative continuous streak method

3.2.2. Liquid culture

- Before starting, make sure that the culture flasks **(Figure 9)** are sterilized as described in **SOP-NAT-STER**.
- Select the liquid media based on the requirements of bacteria that will be cultured.
- Transfer 25-50 mL (volume can be adjusted based on the need) liquid media into a sterile flask in sterile conditions.
- Transfer the sample into the flask. The sample can be a single colony from an agar plate using a sterile inoculation loop or a defined volume of another liquid culture using a sterile pipette tip based on the experimental design.
- Cover the flask with sterilized aluminum foil or cotton.
- Incubate the flask at 36±1°C for a defined time and shaking speed according to the experimental design.
- Always use a sterile media control to make sure that there is no contamination issue.
- Always consider specific atmospheric conditions that the bacteria need to grow. Use an anaerobic jar system and pouches for the desired atmospheric conditions, if needed, according to the manufacturer`s recommendations.

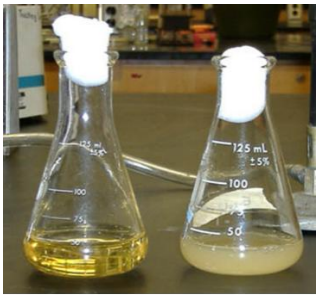

Figure 9. Liquid bacteria culture flasks.

3.3. Storage of bacteria

Each bacterial strain used in Nucleic Acids Technology laboratories should be stored for further use. For this purpose, Microbank Bacterial Preservation System is used according to the manufacturer`s recommendations. Before storage, a single colony obtained using the quadrant streak method should be propagated in another agar plate to make sure about the purity of the bacteria culture.

- Using a sterile cotton swab, collect as much as bacteria colonies from an agar plate of pure culture.
- Place the cotton swab into the storage vial with media and release as much bacteria as possible by rotating the swab in the vial.
- Discard the swab carefully.
- Close the screwed cap of the vial and vortex.
- Discard all the media in the vial leaving only the beads inside.
- Label and store the vial at -80°C until use.
- To propagate the bacteria when needed, take one bead using a sterile tweezer and inoculate onto agar plate as described above.

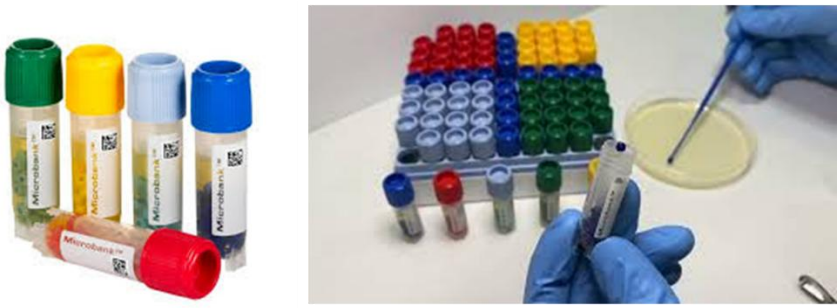

Figure 10. Microbank Bacterial Preservation System

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology laboratories staff on cultivating bacteria in Nucleic Acids Technology laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer

- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- µL – Microliter
- mL – Mililiter

6. Responsibilities

This SOP was prepared by Student name, revised by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- High precision analytical balance
- Metal measuring spoons
- Weighing boats
- Glass bottles
- Ultrapure water system
- Autoclave
- Autoclave gloves
- BSL-2 biosafety cabinet
- Petri dishes
- Glass flasks
- Aluminum foil
- Inoculation loops
- Bunsen burner
- Gloves
- Pipettes
- Pipette tips
- Incubator
- Bacteria storage vials
- Cotton swabs
- Tweezers
- Anaerobic jar system
- Gas pouches

8. Caution/Warning

- Use BSL-2 biosafety cabinet for all the work that involves bacteria.
- Use PPE according to **SOP-NAT-BSL2**.
- Clean the working area before and after the work.

9. References

- Dariel Burdass, John Grainger & Janet Hurst. Basic Practical Microbiology - A Manual, 2018. Published by the Microbiology Society, UK (ISBN 0 95368 383 4).

10. Version Control

Original version of the **SOP-NAT-CULT.v01**. Check **Table 1**.

11.Attachments/Appendices

Appendix I:

- Bunsen burner manual
- Microbank Bacterial Preservation System manual
- Autoclave manual
- Lab scale manual
- BSL-2 biosafety cabinet manual